
C-STGB Real-Time Financial Surveillance Architecture & Conformal Triage Pipeline
1. Streaming Ingest
SWIFT / bKash / Kafka
Apache Arrow Buffers
DuckDB Micro-Batches
12-D Causal Invariants:
Phi_flow, Hawkes lambda_u
s_fwd, s_bwd, [mu_A, sigma_A^2]
Single-event: 0.11 ms
2. C-STGB Neural Core
3-Band Harmonic Temporal GNN
+ Context Edge Gate g_ij
Prunes Chaff Camouflage
(delta_floor >= 0.10, 92.1% Ret.)
Latent GraphSMOTE C_k
Evidence Adaptive Fusion alpha_u
89.60% Cold-Start Minority F1
Cost-Sensitive Head:
tau* = 0.18 (15:1 Cost Ratio)
Inference: 0.16 ms
Class-Cond. CRC Triage
Γ = {1}
Tier 1: High-Risk Quarantine
Immediate Quarantine Hold | Auto FinCEN SAR Package
18 alerts / 10,000 tx (0.18%) | SHA-256 Merkle Audit Receipt
Γ = {0, 1}
Tier 2: Boundary Review Desk
Abstention Set: Escalate to Senior Compliance Examiner
50 alerts / 10,000 tx (0.50%) | 4-Eyes AML Review Desk
Γ = {0}
Tier 3: Straight-Through Clear
Automated Wire-Speed Release (STP) | Residual Leak <0.12%
9,932 tx / 10,000 tx (99.32%) | Zero Latency Impact
MATHEMATICAL FORMULATIONS, RISK BOUNDS & REAL-TIME SYSTEM PERFORMANCE SLAS
Tri-Band Temporal Decay:
Adaptive Edge Trust:
12-D Flow Invariants:
Class-Conditional CRC:
Triage Dispatch:
Fast-Path Latency SLA: Single-event 0.45 ms < 10 ms SLA | Review Queue: −99.50% Workload Cut | Peak VRAM: 428 MB